



JIYOUN LEE

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Education

Dongduk Women's University

B.S. in Information Statistics

- Double Major in Computer Science
- Total GPA : 3.83/4.5, Major GPA : 3.88/4.5 (Credits taken: 130/130)

Seoul, S.Korea
Mar. 2021 – Feb. 2027 (Expected)

Johannes Gutenberg University Mainz

Exchange Semester (Computer Science)

- Relevant Coursework : Database Systems, Reinforcement Learning

Mainz, Germany
Mar. 2025– Aug. 2025

Research Interests

- Personalized and adaptive recommender systems
- Cross-domain recommendation and cold-start user modeling
- Natural language processing and representation learning for heterogeneous user-item data
- LLM-based preference reasoning, recommendation explanation, and evaluation

Publications

- Lee, J. Y., Park, K. H., & Kim, H. H. "An Adaptive Cross-Domain Recommendation Framework for Structurally Disjoint Domains." Applied Sciences, under review.
- Lee, J. Y., Park, K. H., & Kim, H. H. "Cross-Domain Recommendation and Cold-Start Mitigation via Textual Semantic Mapping and Similar-User Transfer." Proceedings of the Annual Symposium of KIPS 2026 (ASK 2026).
- Ha, S. J., Oh, S. H., Ban, S. J., Lee, J. Y., & Kim, H. H. "A Factor Analysis and Regression-Based Prediction Model of Security Deposit Scam Amount for Preventing Rental Scam." Proceedings of the Annual Symposium of KIPS 2024 (ASK 2024).

Project Experience

Adaptive Cross-Domain Recommendation for Structurally Disjoint Domains

First Author & Undergraduate Researcher

Seoul, S.Korea
Mar. 2026 –Present

- Proposed an adaptive cross-domain recommendation framework for domains without shared users, items, or aligned data structures.
- Designed a shared core with semantic, keyword, and preference signals, complemented by domain-specific adaptive modules.
- Addressed new-user cold-start through hierarchical clustering and similar-user transfer.
- Conducted human and LLM-based evaluation; submitted the manuscript to Applied Sciences.

Personalized volunteer activity recommendation system (Capstone Design & ASK 2026)

Team Leader

Seoul, S.Korea
Sep. 2025 – Dec. 2025

- Developed a cross-domain recommendation system connecting user hobby profiles with volunteer activities through semantic matching.
- Combined semantic similarity, preference compatibility, keyword signals, and similar-user transfer for cold-start recommendation.
- Received the Gold Prize at the 2025 Capstone Design Competition and published a first-author conference paper.

Personalized Fashion Recommendation Using Image and Survey Data

Team Leader

Seoul, S.Korea
Aug. 2024 – Nov. 2024

- Developed a recommendation pipeline combining item metadata, image features, and user survey responses.
- Applied object detection and ResNet18-based classification for fashion image preprocessing and style representation.
- Achieved 92.11% accuracy and a 90.37% F1-score and received an Excellence Award at the 2024 Data Creator Camp.

Individualized Housing Recommendation Service (3rd Open Data Forum)

Team member

Seoul, S.Korea
Apr. 2025 – Jul. 2025

- Developed a personalized housing recommendation system using public data and user lifestyle features.
- Designed a weighted ranking method and was selected as a finalist in the Open Data Forum.

Generative AI Platform for Franchise Risk Prediction

Team Leader

Seoul, S.Korea
Sep. 2025 – Nov. 2025

- Developed an LLM-based chatbot for franchise risk analysis and explanation.
- Built a LightGBM-based time-series forecasting model for sales prediction.
- Applied SHAP for interpretable risk analysis and was selected as a finalist in the 13th Big Contest.

Factor Analysis and Prediction of Security Deposit Guarantee Losses

Team member

Seoul, S.Korea
Sep. 2023 – May. 2024

- Analyzed relationships among housing market indicators and security deposit guarantee incidents using public data.
- Built linear and polynomial regression models to predict annual security deposit guarantee incident amounts.
- Achieved an R^2 of 0.97 with the cubic regression model and published the findings at ASK 2024.

Internships

QuadMiner (R&D Intern)

Seoul, S.Korea
Aug. 2026 – Feb. 2027 (Expected)

- Selected for an R&D internship supporting a government-funded research project.
- Expected to assist with literature reviews, technical prototyping, experimental analysis, and research documentation.

WPP Media Korea (Data Analysis Intern)

Seoul, S.Korea
Jan. 2026 – Feb. 2026

- Data Processing: Analyzed advertising performance datasets for premium clients (LVMH).
- Strategic Reporting: Conducted quantitative competitive analysis and data-driven reporting.

Awards & Honors

Awards

2026.01	Gold Prize, 2025-2 Capstone Design Collaborative Learning Competition,	Dongduk Women's Univ
2025.01	Excellence Project Award, Outlier (Academic Club) Data Visualization Project Showcase	Dongduk Women's Univ
2024.12	Excellence Award, 2024 Data Creator Camp (DCC) Undergraduate Category	K-ICT Big Data Center
2024.06	Encouragement Award, The 2nd Generative AI Ideathon	Wrtn Technologies Inc
2024.01	Excellence Project Award, Outlier (Academic Club) Data Analysis Project Showcase,	Dongduk Women's Univ
2023.08	Excellence Award, The 4th National University Student Sharing & Volunteer Hackathon	Korea Red Cross

Honors

2024	semester High Honors, Awarded for outstanding academic performance (Fall Semester)	Dongduk Women's Univ
2024	Semester High Honors, Awarded for outstanding academic performance (Spring Semester)	Dongduk Women's Univ
2023	Semester High Honors, Awarded for outstanding academic performance (Spring Semester)	Dongduk Women's Univ

ExtraCurricular Activities

Outlier (Academic Data Society)

Data Analyst 6th & 7th

Seoul, S.Korea
Sep. 2023 – Jan. 2025

- Led weekly data analysis projects involving predictive modeling, visualization, and research presentation.

BDA (Big Data Analysis Academic Society)

Data Pre-processing Track

Seoul, S.Korea
Mar. 2024 – Sep. 2024

- Completed an intensive curriculum with weekly study sessions and hands-on assignments.
- Specialized in data cleaning and feature engineering using Python to handle real-world datasets.

Teaching

CJ UNIT (11th Gen), CJ OliveNetworks

SW Education Volunteer

Seoul, S.Korea
Aug. 2024 – Dec. 2024

- Conducted weekly software coding workshops for a full semester to bridge the digital divide for underserved students.

Mathematics Mentor

Ansan Youth Shelter

Seoul, S.Korea
Jul. 2023 – Sep. 2023

- Provided personalized mathematics tutoring and academic mentoring for underprivileged youth at a residential shelter.

Certificate & Skills

Certificate • 2024.09 - SQLD

- Skills
- **Programming** Python, R, SQL, C, Java
 - **Machine Learning** PyTorch, TensorFlow, Keras, scikit-learn
 - **Data Analysis** Pandas, NumPy, Matplotlib, Tableau